



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

ClaimGaurd

Insurance Claims Fraud Prioritization System

The domain of the Project:

DATA ANALYTICS

GROUP -1

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Period of the Project

July 2026 to September 2026



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Declaration

The project titled **ClaimGuard: Insurance Claims Fraud Prioritization System** has been mentored by **Ms. Nithyasri Kannathal EL** ma'am and **Mr. Bhargavesh Dakka** sir, organised by SURE Trust, from February 2026 to September 2026 for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

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Abstract

Insurance companies process claims largely in the order they are received, which means claims with a high likelihood of fraud receive the same review priority as routine, genuine claims. This delays fraud detection, wastes investigator time on low-risk cases, and does not scale as claim volume grows. ClaimGuard is an end-to-end data analytics and machine learning system that scores every insurance claim for fraud risk at the moment it is filed, automatically prioritizing high-risk claims at the top of an investigator's review queue. The system combines exploratory data analysis, a Logistic Regression classification model (benchmarked against Random Forest), a live Streamlit investigator dashboard, a Supabase (PostgreSQL) data layer, an automated n8n alerting workflow, and an executive Power BI reporting layer. Working on a dataset of 15,420 historical insurance claims, the project demonstrates a complete, deployable fraud-triage pipeline - from raw data to an automated, explainable decision support tool.

Introduction and Problem Statement

Fraudulent insurance claims are a significant source of financial loss for insurers. The core operational problem this project addresses is that claims are traditionally reviewed on a first-in-first-out (FIFO) basis - the order in which they arrive - rather than by their actual risk of being fraudulent. This creates three concrete issues:

- A high-risk fraudulent claim can sit in a queue behind several routine, genuine claims, delaying detection and increasing the chance that a fraudulent payout is approved before investigation.
- Investigator time, a limited resource, is spent reviewing claims in arrival order rather than in order of risk, reducing overall investigative efficiency.
- As claim volume grows, a purely manual review process does not scale, and decision consistency across different investigators degrades.

ClaimGuard's central idea is to replace this arrival-order queue with an automated, risk-based queue - scoring every claim the moment it is filed and surfacing an explainable reason alongside the score, so that investigator attention is directed to the claims that matter most.

Objectives

- Build a machine learning model that estimates the probability that a given insurance claim is fraudulent, using only information available at the time the claim is filed (i.e., with zero data leakage).
- Identify the strongest behavioural and situational indicators of fraud through exploratory data analysis.
- Design a single, denormalized database schema that supports fast, join-free querying for both the live application and reporting layer.



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- Deploy a live, usable investigator dashboard that surfaces prioritized claims with a plain-language reason for each flag.
- Automate daily monitoring so that high-risk claims are proactively emailed to investigators without requiring them to check the dashboard manually.
- Provide an executive-level reporting view (Power BI) summarizing claim volume, risk distribution, and fraud drivers for leadership.

Dataset Description

The project uses a dataset of 15,420 historical insurance claims (the fraud_oracle.csv insurance dataset), each with 33 features spanning claim timing, policyholder demographics, vehicle details, accident circumstances, claim history, and policy information, plus a binary ground-truth label, FraudFound_P, indicating whether the claim was confirmed fraudulent.

Metric	Value
Total claims	15,420
Genuine claims	14,497 (~94%)
Fraudulent claims	923 (~6%)
Total features	33 (+ target column)
Data leakage	0% — all features known at filing time

Exploratory Data Analysis Key Insights

- Class imbalance: genuine claims dominate the dataset (94% vs 6%), which directly informed the modeling and evaluation strategy used later in the project.
- Utility Vehicle claims: show the highest fraud rate of any vehicle category, at approximately 11.9% — nearly double the dataset's 6% overall baseline.
- Policyholder Fault: claims where the policyholder was at fault show an elevated fraud rate of approximately 7.8%.
- First-time claimants: show a fraud rate of approximately 7.2%, higher than repeat claimants.
- Missing corroboration: claims with no police report filed or no witness present each show an elevated fraud rate of approximately 6.9%, making these two of the most reliable behavioural fraud indicators in the dataset.



Database Design

Rather than a normalized, multi-table schema, ClaimGuard uses a single denormalized PostgreSQL table, claims, keyed by claim_id. This design choice avoids joins entirely at query time, which keeps both the live Streamlit application and the automation workflow fast and simple, since every feature the model needs is a column in one table.

6.1 Column Groups

Group	Representative Columns
Identifiers	claim_id (PK), PolicyNumber, RepNumber (excluded from ML)
Timing	Month, WeekOfMonth, DayOfWeek, Year, MonthClaimed, DayOfWeekClaimed
Policyholder Profile	Sex, MaritalStatus, Age, AgeOfPolicyHolder, DriverRating
Vehicle	Make, VehicleCategory, VehiclePrice, AgeOfVehicle, NumberOfCars
Accident Details	AccidentArea, Fault, Days_Policy_Accident
Fraud-Relevant History	PastNumberOfClaims, PoliceReportFiled, WitnessPresent, AddressChange_Claim
Policy Details	PolicyType, BasePolicy, AgentType, Deductible, Days_Policy_Claim
Target	FraudFound_P — ground-truth label used for training

PolicyNumber and RepNumber are retained in the table for human reference (e.g. looking up a policy or agent) but are explicitly excluded from the feature set used to train the model, since they are identifiers rather than genuine risk signals.



Feature Engineering

- Dropped non-predictive identifier columns (PolicyNumber, RepNumber) prior to model training.
 - Encoded categorical fields (e.g. Fault, Sex, VehicleCategory, PolicyType) into numeric form for model compatibility.
 - Validated the full feature set against the claim-filing timeline to confirm zero data leakage — no feature reflects information only available after investigation or claim closure.
-

Machine Learning Methodology

8.1 Models

Two classification models were built and compared:

Model	Role	Rationale
Logistic Regression	Primary model	Interpretable coefficients let each risk factor's contribution to a claim's score be explained directly to an investigator.
Random Forest	Comparison model	Higher-capacity, non-linear model used to validate whether the simpler baseline was missing meaningful patterns.

8.2 Handling Class Imbalance

With fraud representing only ~6% of claims, a model optimized naively would achieve high accuracy while catching almost no fraud. SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data to generate synthetic examples of the minority (fraud) class, giving the model a more varied and balanced set of fraud examples to learn from, rather than relying solely on class-weighting.

8.3 Threshold Tuning

The default classification threshold of 0.5 was deliberately lowered to 0.4. This reflects a conscious precision-recall trade-off: missing a genuine fraud case is more costly to the business than an investigator spending extra time reviewing an additional false positive, so the model was biased toward higher recall.



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8.4 Evaluation Metrics

Because of the class imbalance, model performance was judged on Precision, Recall, and F1-score rather than raw accuracy, which would be misleading on a 94/6 imbalanced dataset. Models were evaluated on a held-out test split not seen during training.

Model Architecture: Training vs. Inference

9.1 Training Pipeline (Offline)

Raw Claims Data to Feature Engineering to SMOTE + Train/Test Split to Train Models (Logistic Regression + Random Forest) to Save Trained Model.

This pipeline runs once (and can be re-run periodically as new labeled data becomes available); it produces a serialized, trained model artifact.

9.2 Inference Pipeline (Live)

New Claim Filed to Trained Model Scores It to Score Saved to Supabase to Shown in Dashboard + Automated Alerts.

Every new claim is scored instantly using the already-trained model - no retraining occurs at inference time - and the resulting score and explanation are persisted to the database for both the live dashboard and the reporting layer to consume.

9.3 Per-Claim Scoring Mechanics

At the level of a single claim, scoring proceeds as follows: (1) the claim's 33 features are captured at filing time; (2) categorical fields are encoded numerically; (3) each feature is multiplied by its learned Logistic Regression weight; (4) the weighted values are summed and passed through a sigmoid function, producing a probability between 0 (genuine) and 1 (fraud); (5) if that probability exceeds 0.4, the claim is flagged high-risk and prioritized in the queue.



System Architecture and Technology Stack

Layer	Technology	Purpose
Application	Streamlit (Python)	Investigator-facing dashboard: live overview, Claim ID search, sorted risk queue.
Database	Supabase (PostgreSQL)	Stores the single claims table and all scored results.
Automation	n8n	Scheduled daily workflow that queries and emails high-risk claim alerts.
Reporting	Power BI	Executive dashboard: KPI cards, risk distribution, monthly trends, fraud-driver breakdown.
Modeling	scikit-learn (Python)	Logistic Regression & Random Forest training and evaluation.

Automation Workflow (n8n)

A scheduled n8n workflow runs daily at 8:00 AM and performs four steps in sequence:

- Schedule Trigger — fires automatically every day at 8 AM (manually re-triggerable for demonstration purposes).
- Query Supabase — fetches all currently scored claims from the claims table.
- Filter High-Risk — retains only claims scoring above 0.7 — a deliberately higher bar than the 0.4 classification threshold, so that proactive email alerts are reserved for only the most urgent cases.
- Send Email Alert — emails the filtered, consolidated list directly to investigators.

Power BI Reporting Layer

- KPI Cards: Total Claims, High-Risk Claims, and Average Risk Score.
- Risk Level Distribution: a donut chart showing the Low / Medium / High risk split across the current claim volume.
- Claims by Month: a trend chart of claim volume across the year.



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- Why Claims Get Flagged: a bar chart of the real, dataset-derived fraud rate for each risk signal (e.g. Utility Vehicle, Policyholder Fault, No Police Report).

Results and Business Impact

- Operational impact: of the 15,420 claims in the current queue, approximately 4,736 are flagged high-risk, concentrating investigator attention on a genuine, actionable subset rather than the full volume.
- Explainability: every flagged claim is accompanied by a plain-language reason (e.g. “no police report was filed, the policyholder was at fault, and no witness was present”), making the system's output defensible in an audit or compliance review.
- Business value: earlier detection reduces fraudulent payouts, investigator time is spent where it matters most, decisions are auditable, and the system's automated design scales with claim volume without a proportional increase in manual effort.

Limitations

- The dataset used is a static historical snapshot rather than a live, continuously updated feed, so the automation workflow is demonstrated as an on-demand triggered run rather than a true daily production feed.
- The model is trained on one dataset's fraud patterns and geography; it would require retraining and revalidation before being applied to a different insurer's data or region.
- The current threshold (0.4) and alert bar (0.7) were chosen analytically rather than tuned against live investigator feedback.

Future Scope

- Introduce model monitoring to detect performance drift as new claims accumulate over time.
- Support periodic retraining on a rolling window of newly labeled claims.
- A/B test the classification threshold and alert bar against real investigator outcomes rather than fixed values.
- Extend the automation workflow to a genuinely continuous data feed rather than a static snapshot.



Conclusion

ClaimGuard demonstrates a complete, end-to-end data analytics and machine learning pipeline for insurance fraud prioritization - from raw claims data and exploratory analysis, through an interpretable, imbalance-aware classification model, to a live, explainable investigator dashboard supported by automated daily alerting and executive reporting. By replacing a first-in-first-out review process with an automated, risk-ranked queue, the system directly addresses the core inefficiency in traditional claims handling: that investigator attention is spent on arrival order rather than actual risk.

Tools and Technologies

- Python (Pandas, scikit-learn)
- SQL
- PostgreSQL (Supabase)
- Streamlit
- Power BI
- n8n
- Git/GitHub



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